Topic 12 — Human Geography (Settlements)

> **Sources baked in:** NCERT Class 12 Fundamentals of Human Geography + Human Settlements, PWOnlyIAS / Drishti notes, Census town definitions, MoHUA / MoRD schemes, UPPCS Prelims 2018–2025 **Weight:** ★★★ — village types, town functions, Smart City lists (UP), scheme years, metro vs million city; also teach nature/scope even if not yet asked **Last verified:** August 2026 **Current Affairs:** Smart Cities Mission (25 Jun 2015); AMRUT 2.0 (1 Oct 2021); SPMRM **launch 2016** (not 2015); ISAC-2020; UP **10 Central** Smart Cities

Current Affairs (this topic)

Year	Fact	Why asked	Source
25 Jun 2015	Smart Cities Mission + PMAY-U same day	Year pair vs JNNURM 2005	MoHUA
Jun 2015	AMRUT (~500 cities)	2025 chronology	MoHUA
1 Oct 2021	**AMRUT 2.0** — water-secure towns	Successor of AMRUT	PIB / MoHUA
21 Feb 2016	**SPMRM** launched (not “2015 launch”)	2023 Q143	MoRD
ISAC-2020	Indore/Surat best cities; **UP best state**; category winners above	2022 Q26	PIB / MoHUA
100 / 10	100 Smart Cities nationally; **10 Central in UP**	2018 + 2020	MoHUA
2019	UP **State** Smart Cities (7 Nagar Nigams)	Does **not** change 2018/2020 keys of 10	UP DoUD
Mission window	Smart Cities Mission timeline extended to **31 Mar 2025**	Completing works	MoHUA

Million-city **population (जनसंख्या) ranks**. Missions and settlement types

N.0 Framework — Nature & Scope of Human Geography

Human geography studies the **relationship between people and the physical earth**, the **spatial distribution** of human phenomena, and social–economic differences across places.

- Physical geography studies nature; human geography studies how societies live in, use and reshape that nature.
- The two are linked: humans naturalise themselves to the environment and also humanise nature through farms, houses, roads and cities.

Schools / approaches

Approach	Core fact	Name pair
Environmental determinism	Nature controls human life and culture	**Friedrich Ratzel**; **Ellen C. Semple**
Possibilism	Nature offers possibilities; humans choose	**Vidal de la Blache**
Neo-determinism / stop-and-go	Nature sets limits; man can proceed carefully	**Griffith Taylor**

Branches / fields

- Population geography, settlement geography, economic geography, social geography, political geography and cultural geography are standard subfields.
- This chapter's home is mainly **settlement geography** — rural patterns, urban hierarchy, town functions and urban missions — but nature/scope can still be asked as a fresh angle.

Settlement vocabulary that opens every rural–urban stem

- A **settlement** is an organised place where people live and work.
- **Site** is the immediate ground (water, slope, soil, flood-free height).
- **Situation** is the place in the wider region (routes, markets, other towns).
- Settlements may be temporary (camps, seasonal herding) or permanent (villages, towns, cities).

N.1 Settlements — Rural vs Urban

Type of settlement is read from **built-up form and inter-house spacing**, not from a vague "developed vs backward" label.

- **Wet-point** settlements cluster **around water** (well / oasis) — common compact Rajasthan pair.
- **Dry-point** settlements sit on **flood-free upland** or a levee in wet plains.

Feature	Rural	Urban
Economy	Mainly primary (farm, forest, fish, village-scale mine)	Secondary / tertiary / quaternary
Density / size	Lower; hamlet to village	Compact; town to mega city
Link	Supply food and raw (रॉ) material	Supply goods, services and jobs
Census status	Village / rural	**Statutory town** or **census town**
Social life	Face-to-face primary groups	Secondary groups and specialised jobs

- A place is Census-urban if it is a **statutory town** (municipality, corporation, cantonment or notified town area) **or** a **census town**.
- **Census town** needs all three together: population ≥ **5,000**, density ≥ **400**/km², and ≥ **75%** of **male main** workers in **non-agriculture**.
- Census **2011** urban share is **31.16%** (~**37.7 crore** people; about **7,935** towns including statutory + census).

- Urbanisation percentage is urban population divided by total population, times 100.
- River-bank trap: Agra (Yamuna (यमुना)), Patna (Ganga (गंगा)), Kolkata (Hooghly (हुगली)) and Varanasi/Prayagraj (Ganga) are river towns; **Bhopal is not** on a major river bank.
- A large village is **not** urban unless notified or it meets the census-town triple.

N.2 Types of Villages

Rural settlement pattern differs for three NCERT reasons.

- **Physical factors** include terrain, altitude, climate and especially **water**.
- **Cultural / ethnic factors** include caste and religion.
- **Security / defence** also shapes compact clustering in some belts.

NCERT four-fold rural pattern

Type	Form	Where (India) — match this
Clustered / nucleated / agglomerated	Compact houses with living area **separate** from fields; streets may be rectangular, radial or linear	Fertile **alluvial (जलोढ़) plains**; also **NE plains**; **Rajasthan** (थार) (scarce water → cluster at well/oasis); **Bundelkhand + Nagaland** (defence)
Semi-clustered / fragmented	One main cluster plus groups pushed a little away	**Gujarat** and **Rajasthan** fragmentation. The dominant / land-owning community sits at the **centre**; lower strata / menial workers sit on the **outer flanks**
Hamleted	Several physically separated units, **one village name**	**Middle & lower Ganga**, **Chhattisgarh**, lower **Himalayan valleys**
Dispersed / isolated	Isolated huts / tiny hamlets on hills, jungle, pasture slopes	**Meghalaya, Uttarakhand, Himachal, Kerala**; forests & hills of **North-East**

Hamlet local names (one village, many units)

Name	Region
Panna / para / palli	East / Ganga–Chhattisgarh belt
Nagla	Western UP / Ganga plain
Dhani	Rajasthan

Shape add-ons (not a fifth NCERT type)

Shape	Note
Linear	Road, river, canal, coast, levee
Rectangular	Planned / northern plains field grid
Circular / radial	Around tank, temple, well

Shape	Note
Star / T / Y	Where routes meet

Rural house-material pairs (climate)

Belt	Note
Heavy rain (W. coast, NE)	**Sloping** roofs; bamboo / wood in NE
Arid Rajasthan	**Flat** roofs; compact around water
Himalaya (हिमालय)	Stone / slate
Ganga plain	Mud / brick; courtyard houses

> **Logic:** Do not **key "Thar = dispersed." NCERT: Rajasthan is** clustered **because of water scarcity. Dispersed NCERT option =** forests and hills in the North-East**, not the Ganga plain.

N.3 Functional Classification of Towns

Size (Census class) and function are **different** questions.

- No town is purely one-function; classify by the **dominant** job.
- As a city becomes a metropolis, functions intertwine — Mumbai is port (बंदरगाह) + trade (पण्याध्यक्ष) + industry + finance together.

Census size classes (2011)

Class	Population	2011 fact
I (city)	≥ **1,00,000**	**468** towns; ~**60%** of India's urban population
II	50,000–99,999	
III	20,000–49,999	
IV	10,000–19,999	
V	5,000–9,999	
VI	< 5,000	

Dominant function — NCERT example list (learn the match)

Type	Example
Administrative	Chandigarh, New Delhi, Bhopal, Shillong, Guwahati, Imphal, Srinagar, Gandhinagar, Jaipur, Chennai, **Lucknow** (लखनऊ)

Type	Example
Industrial	Jamshedpur, Bhilai, Durgapur, Modinagar, Barauni, Sindri, Mumbai, Madurai, **Kanpur** (कानपुर)
Transport / port	Kandla, Kochi, Visakhapatnam, **Mughalsarai / Pt. Deen Dayal Upadhyaya Jn.**, Itarsi, Katni
Commercial	Kolkata, **Saharanpur**, Satna
Mining	Raniganj, Jharia / Dhanbad, Digboi, Ankleshwar, **Singrauli**
Garrison / cantonment	Ambala, Jalandhar, Mhow, Babina, Udhampur, **Meerut** (मेरठ)
Educational	Roorkee, **Aligarh**, Pilani, Varanasi (BHU), Prayagraj
Religious / cultural	**Varanasi** (वाराणसी), Mathura, Amritsar (अमृतसर), Madurai, Puri (पुरी), Ajmer (अजमेर), Pushkar, Tirupati, Kurukshetra, Haridwar (हरिद्वार), Ujjain (उज्जैन)
Tourist	Nainital (नैनीताल), Mussoorie (मसूरी), Shimla, Pachmarhi, Jodhpur, Jaisalmer, Udhagamandalam (Ooty), Mount Abu, **Agra** (आगरा)

Urban evolution (NCERT three layers)

Layer	Fact
Ancient (~2000+ yrs)	**Varanasi, Prayag (Prayagraj), Pataliputra (पाटलिपुत्र) (Patna), Madurai** — religion / river / administration
Medieval (~100 towns)	Fort / court HQ — **Delhi, Hyderabad, Jaipur, Lucknow, Agra, Nagpur**
Modern — European foothold	First **ports** at Surat, Daman, Goa (गोवा), Puducherry; later **Mumbai, Chennai, Kolkata**; hill stations as summer resorts; cantonments
After Independence	New capitals such as **Chandigarh, Bhubaneswar, Gandhinagar, Dispur**; industrial towns such as **Durgapur, Bhilai, Sindri, Barauni**
Satellite towns	Old towns pulled into a metro's orbit — **Ghaziabad, Gurugram, Rohtak** around Delhi; Navi Mumbai / Thane around Mumbai

Planned / new-town architects (extras)

City	Note
New Delhi	Lutyens / Baker
Chandigarh	**Le Corbusier**
Bhubaneswar	Otto Koenigsberger
Jamshedpur	Company (कंपनी) / steel town (Tata (तत))

N.4 Metropolitan Cities & Urban Hierarchy

Term	Fact (Census / NCERT)
Town	Urban centre below Class I, or loosely any urban place
City / Class I	≥ **1 lakh**
Million city	≥ **10 lakh** · Census 2011 = **53** UAs · largest **Greater Mumbai** · smallest **Kota**
Metropolitan (NCERT)	**1–5 million** (10 lakh–50 lakh)
Mega city (NCERT table)	**> 5 million** · **six** in 2011: Greater Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad · ~**21%** of urban pop
Megacity (UN)	≥ **10 million** · 2011 India: Mumbai, Delhi, Kolkata
Conurbation	Once-separate cities fused (Mumbai–Thane)
Megalopolis	Chain of conurbations (Boswash type)
Primate city	One city dwarfs the national urban system — **India does not** have a single primate city (Mumbai vs Delhi)
NCR	Delhi + **Ghaziabad, Noida, Greater Noida, Meerut** (UP) + Gurugram, Faridabad (HR)

Urban agglomeration — any one of three

1. A town + its adjoining **outgrowths**
2. Two or more **contiguous towns**, with or without OGs
3. A city + one or more adjoining towns + their OGs, as one contiguous spread **Outgrowth (OG) examples:** railway colony (उपनिवेश), university campus, port area, military cantonment — outside statutory limits but contiguous. **2011 UA size order (top six)Greater Mumbai > Delhi > Kolkata > Chennai > Bengaluru > Hyderabad** (Older NCERT MCQ used **2001** order Mumbai > Kolkata > Delhi > Chennai — do not mix.) UP million+ **years** and UA order Kanpur > Lucknow > Ghaziabad > Agra.

> **Logic:** ** Class I (**1 lakh**) ≠ metro (**10–50 lakh**) ≠ NCERT mega (**>50 lakh**, six cities) ≠ UN megacity (**1 crore**).

N.5 Smart Cities & Heritage Cities

The **Smart Cities Mission** launched on **25 June 2015** under **MoHUA** for **100** cities. Each city implements the Mission through a city-level **SPV** under the Companies Act. The strategy pairs **Area-Based Development** (retrofit, redevelopment or greenfield) with **pan-city** ICT.

UP's 10 Central / Mission cities (2020 paper count is **10**) — learn the **round** as well as the list:

Round	When	Cities
Fast-track / first UP entry	**May 2016**	**Lucknow** (first UP city on the Mission list)

Round	When	Cities
Round 2	**Sep 2016**	**Kanpur, Agra, Varanasi**
Round 3	**Jun 2017**	**Prayagraj (Allahabad (इलाहाबाद)), Aligarh, Jhansi**
Round 4	**2018**	**Moradabad, Bareilly, Saharanpur**

Not in that ten (and **not** in Mission-100): **Ghaziabad**, Noida, Meerut, Rae Bareli.

Area-Based Development (ABD) terms

- **Retrofit** improves services and infrastructure in an **existing** built-up area without wholesale replacement.
- **Redevelopment** replaces an existing built-up pocket with a new layout and buildings.
- **Greenfield** develops a previously vacant or unused site as a new urban pocket.
- **Pan-city** adds ICT and smart solutions that serve the **whole city**, not only one ABD pocket.
- Centre:State funding for the Mission is typically **50:50** through the city SPV.

UP State Smart Cities (2019) — CM extension to remaining Nagar Nigams; **not** the Central Mission list: Ayodhya, Firozabad, **Ghaziabad**, Gorakhpur, Mathura, Meerut, Shahjahanpur. Prelims 2018/2020 still key **10**. **ISAC-2020** (awards presented Apr 2022):

Category	Winner in the paper's list
Culture	**Indore**
Governance	**Vadodara**
Social Aspects	**Tirupati**
Urban Environment	**Bhopal**

Best cities in that cycle include **Indore** and **Surat**. The best **state** is **Uttar Pradesh**. Varanasi also won leadership / COVID / water awards in that cycle. **HRIDAY** (National Heritage City Development and Augmentation Yojana, **Jan 2015**, MoHUA) covers **12** cities nationally. In **UP** only **Varanasi** and **Mathura** (मथुरा) are in HRIDAY — not Prayagraj or Ayodhya. National 12 (for NOT-matched): Ajmer, Amravati, Amritsar, Badami (बादामी), Dwarka (द्वारका), Gaya (गया), Kanchipuram (कांचीपुरम), **Mathura**, Puri, **Varanasi**, Velankanni, Warangal (वारंगल).

> **Logic:** ** Ghaziabad as a "Smart City" in 2018/2020 keys is **false**. State-2019 tag does not rewrite those papers.

Inline PYQ — UPPCS Prelims 2018, Q41

Which of the following cities in Uttar Pradesh has NOT been selected for development under the Smart City Development Programme of the Central Government?

A. Lucknow

B. Allahabad

C. Ghaziabad

D. Kanpur

Show answer

Ans: C — Ghaziabad is outside the Central Mission-100 / UP Central-10 list.

Inline PYQ — UPPCS Prelims 2020, Q11

No. of cities in U.P. which are covered under the 'Smart City' Scheme

A. 10

B. 12

C. 15

D. 18

Show answer

Ans: A — UP Central / Mission count = **10**.

N.6 Smart Villages & Scheme Years

Smart village / rurban means urban-level amenities **without** wiping village social life. The nodal ministry is **MoRD**, not MoHUA.

Scheme	Year / fact	Trap
SPMRM (Shyama Prasad Mukherji Rurban Mission)	Launch **21 Feb 2016** (Cabinet **16 Sep 2015**); **MoRD**; about **300** clusters	**2015** is **not** the launch year (2023 NOT-matched)
Sansad Adarsh Gram Yojana (SAGY)	**11 Oct 2014**	
DDU-GKY	**2014**	
Rural Housing Interest Subsidy Scheme	**2017** (that paper)	
JNNURM	**2005**	Confusing with AMRUT
National Urban Housing and Habitat (वास स्थान) Policy	**2007**	
AMRUT	**June 2015**; about **500** cities (all 1-lakh+ ULBs, capitals, HRIDAY cities and selected others)	2005
AMRUT 2.0	**1 Oct 2021** — water security; all statutory towns	Calling it 2015
Jal Jeevan Mission	**2019** (Har Ghar Jal)	Putting it before AMRUT

Scheme	Year / fact	Trap
PMAY-U	**25 Jun 2015** (same day as Smart Cities) — housing neighbour, year fact only	
SBM-Urban	**2 Oct 2014** (with rural SBM) — sanitation neighbour	Calling it 2015 Smart Cities
DAY-NULM	Urban livelihoods (MoHUA)	Mixing with SPMRM / DDU-GKY

- **Rurbanization** in the 2022 sociologist set is locked to **G.S. Ghurye**.
- Do not mark M.N. Srinivas (Sanskritization), Yogendra Singh (modernisation) or Spencer (evolutionism) for that stem.
- **Global Village** (McLuhan) rests on **transport and communication** development — not "UN created it."

Scheme chronology (2025 paper order)

Scheme	Year
JNNURM	**2005**
National Urban Housing and Habitat Policy	**2007**
AMRUT	**2015**
Jal Jeevan Mission	**2019**

> **Logic:** ** AMRUT ≠ JNNURM. Rurban **launch 2016**. Smart village ≠ Smart City Mission.

Inline PYQ — UPPCS Prelims 2023, Q143

Which of the following (Scheme — Year) is not correctly matched?

A. DDU-GKY — 2014

B. Rural Housing Interest Subsidy Scheme — 2017

C. SPMRM — 2015

D. SAGY — 2014

Show answer

Ans: C — SPMRM was **launched in 2016** (Cabinet approval 2015 is not the launch-year fact).

Inline PYQ — UPPCS Prelims 2025, Q72

Arrange the following in chronological order of launch:

1. AMRUT
2. JNNURM

3. Jal Jeevan Mission
4. National Urban Housing and Habitat Policy

A. 2, 4, 1, 3

B. 1, 2, 3, 4

C. 2, 1, 4, 3

D. 4, 2, 1, 3

Show answer

Ans: A — 2005 → 2007 → 2015 → 2019.

Inline PYQ — UPPCS Prelims 2022, Q (rurbanization)

Who among the following is associated with the concept of rurbanization in the standard UPPCS option set?

A. M. N. Srinivas

B. Yogendra Singh

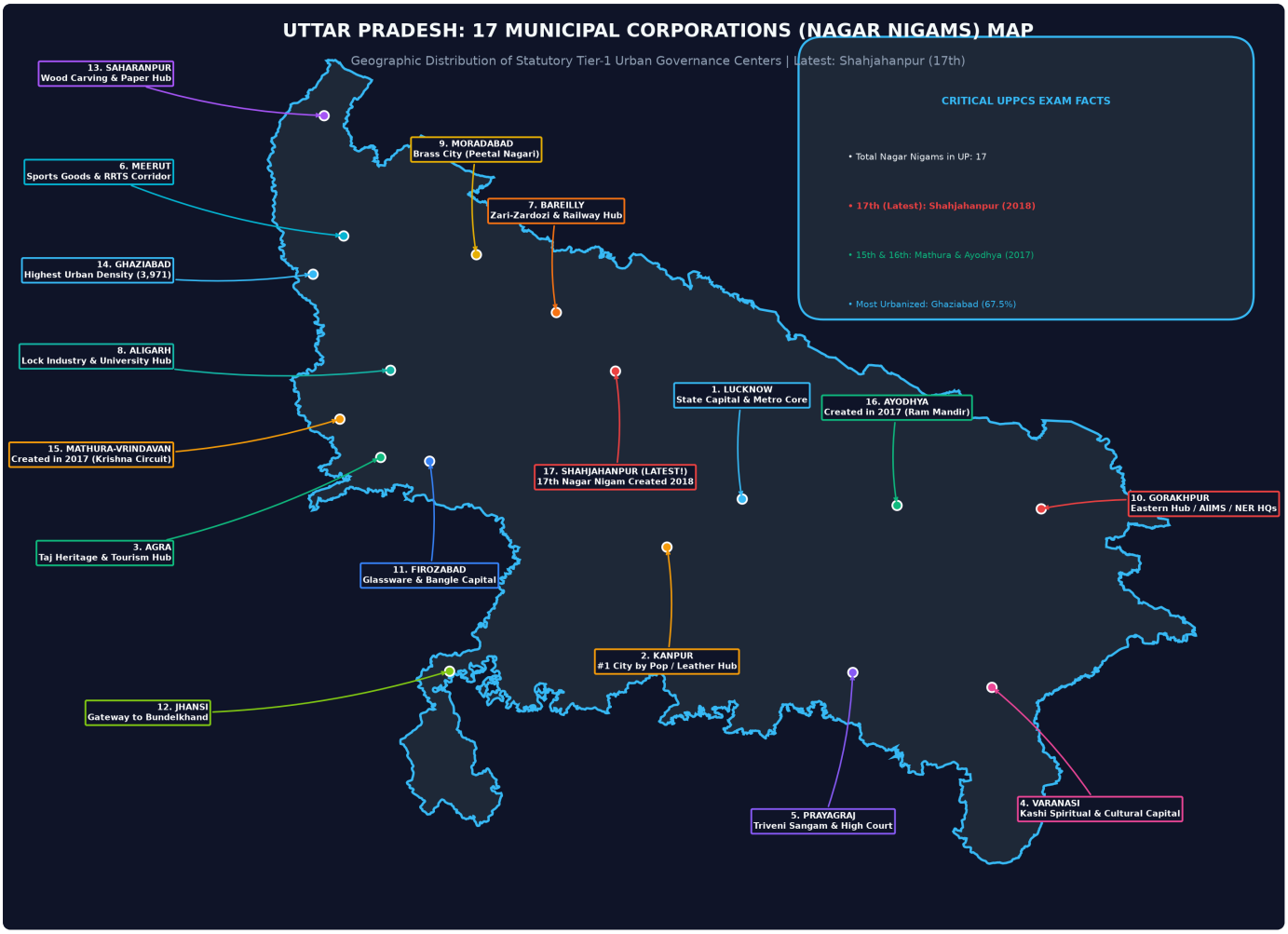
C. G. S. Ghurye

D. Herbert Spencer

Show answer

Ans: C — **G.S. Ghurye**.

N.7 Uttar Pradesh Focus — Urban Centres



City	Functional / tag
Lucknow	State capital — administrative; first UP Smart City (May 2016)
Kanpur	Industrial / leather-textile; first UP million city (**1971**)
Prayagraj (प्रयागराज)	Ancient Prayag; educational; Smart City round 3; **not** HRIDAY
Varanasi	Ancient; religious-educational (BHU); HRIDAY; Smart City round 2
Agra	Medieval; tourist (Taj); Smart City round 2
Aligarh	Educational (AMU); industrial neighbour of the western UP belt; Smart City round 3
Meerut	Garrison / NCR satellite — **not** Central Smart-10
Ghaziabad	NCR satellite / large UA — **not** Central Smart-10 (State 2019 only)
Noida / Greater Noida	Planned satellite of Delhi — **not** Mission-100
Saharanpur	NCERT **commercial** town; Smart City round 4
Bareilly / Moradabad / Jhansi	Smart City round 4 / 3; Moradabad brass
Mathura	Religious; **HRIDAY**; State Smart City 2019 (not Central-10)

City	Functional / tag
Ayodhya (अयोध्या)	Religious; State Smart City 2019; **not** HRIDAY
Singrauli belt (Sonbhadra (सोनभद्र))	Mining / energy belt — mining-town type
Mission fact	Fact
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Central Smart Cities	**10** — Ghaziabad is **out** of Mission-100
State Smart Cities (2019)	7 Nagar Nigams — do **not** change 2018/2020 keys
HRIDAY	**Varanasi** and **Mathura** only
ISAC-2020	UP is the **best state**
NCR (UP slice)	Ghaziabad, Noida, Greater Noida, Meerut